The new token-based authentication method is working well for your mobile and desktop apps, but your database administrators are worried that the tokens table keeps growing larger without any tokens ever being removed. This also creates a potential DoS attack vector, because an attacker could keep logging in to generate enough tokens to fill the database storage. You should implement a periodic task to delete expired tokens to prevent the database growing too large. This is a one-line task in SQL, as shown in listing 5.6. Open DatabaseTokenStore.java and add the method in the listing to implement expired token deletion.

```
Listing 5.6 Deleting expired tokens
public void deleteExpiredTokens() {
    database.update(
        "DELETE FROM tokens WHERE expiry <
current_timestamp");    #A
}
```

#A Delete all tokens with an expiry time in the past

To make this efficient, you should index the expiry column on the database, so that it does not need to loop through every single token to find the ones that have expired. Open schema.sql and add the following line to the bottom to create the index:

```
CREATE INDEX expired_token_idx ON tokens(expiry);
```

Finally, you need to schedule a periodic task to call the method to delete the expired tokens. There are many ways